Rank Instability and Representation Repair for Automatic Modulation Classification under Structured Interference

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***Abstract*—Learned wireless receivers trade information-rich signal representations for compact front ends under tight inference budgets, but their relative ranking can change across interference and mobility regimes. In a source-disjoint simulation campaign, a sealed whole-configuration contrast improves macro-F1 by 4.57 percentage points, while isolated teacher and route-count effects remain unresolved. A post-hoc severe-condition analysis reveals a compact-versus-I/Q ranking reversal inside the campaign, but an independently generated severe-held evaluation shows that neither this favorable ordering nor a campaign-derived overlap–SIR diagnostic surface transports uniformly. On the same frozen severe-held benchmark, a lightweight received-I/Q sidecar produces a consistent 4.37 percentage points relative repair at about 3% additional MACs. A prospectively frozen retrained ablation with the I/Q input permanently zeroed yields -0.54 percentage points relative to the spectral baseline, supporting attribution of the repair to received-I/Q information rather than to added capacity or fusion structure. All models remain low in absolute macro-F1 at this extreme operating point, so the result is a relative robustness repair rather than deployable accuracy. These results frame AMC model comparison as a learned-receiver robustness problem: model ranking is regime-dependent, campaign-derived condition surfaces may fail to transport, and preserving a small information-rich signal path can recover part of the exposed degradation.**

***Index Terms*—Automatic modulation classification, structured interference, rank instability, representation repair, domain shift, compact neural networks, interference overlap, vehicular Doppler.**

Evidence status. The sealed family resolves one positive contrast, A5 versus A0; the isolated teacher-form, route-count, and clean-retention gates do not pass. All campaign-diagnostic, severe-held, repair, and control results are exploratory post-campaign evidence and do not alter the sealed family.

I. Introduction

Automatic modulation classification supports spectrum monitoring and blind receiver configuration by inferring a waveform's modulation from received samples. Deep AMC has progressed from convolutional I/Q classifiers [1], [2] to complex-valued, multi-stream, recurrent, and Transformer-like representations [3]–[7]. Compact architectures remain attractive when inference cost matters [8], [9], but average accuracy alone does not say where a compact representation is preferable.

This omission is consequential in mobile and vehicular receivers. Structured interference and mobility can change which representation is preferred under a fixed compute budget. A marginal score then hides both the rank changes and the risk that a campaign-internal favorable region will fail under a new regime. The issue extends beyond AMC model comparison. Learned wireless receivers often trade information-rich signal representations for compact front ends under tight inference budgets, while their deployment environments span channel, mobility, and interference regimes not represented by one training campaign. A static leaderboard is therefore insufficient whenever the preferred representation changes with operating condition. The more relevant design questions are whether such ranking relationships transport and whether lost robustness can be recovered without reverting to a full high-cost receiver.

This paper asks whether compact spectral and high-capacity I/Q rankings remain stable across structured-interference conditions. It then tests whether an apparent campaign-internal boundary survives independent severe-domain validation. When that interpretation fails, we ask whether the degradation can be diagnosed and repaired at the representation level. We answer using a frozen source-disjoint simulation campaign, artifact-only analysis, and prospective interventions. The base model, VIMD-Net, maps a complex STFT to three allocation routes and uses a training-only simulation-component teacher. No sealed contrast confirms either element individually. The contribution is the rank-instability evidence, its independent cross-regime transport test, and the representation-repair chain.

The contributions are:

- a controlled sealed A5–A0 whole-configuration contrast of 4.57 percentage points ($[3.65, 5.49]$ percentage points), while the isolated teacher-form and route-count effects remain unresolved; the result is not interpreted as a component-level or capacity-independent effect;
- a full SNR–SIR study showing that marginal model ordering can reverse inside the frozen campaign, motivating explicit ranking-stability analysis;
- a source-disjoint independently generated severe-domain evaluation showing that the campaign-internal favorable ranking and its low-dimensional condition surface do not transport uniformly across unseen jammer, mobility, and channel combinations; and
- on the same frozen severe-held benchmark, a lightweight received-I/Q sidecar provides a consistent 4.37 percentage points relative repair at about 3% additional MACs. Tested spectral width scaling does not reproduce the repair, and a prospectively frozen retrained zeroed-I/Q ablation supports attribution to received-I/Q information rather than to added capacity or fusion structure.

The claim is bounded accordingly. Compact spectral and high-capacity I/Q representations exhibit condition-dependent rank instability inside the frozen campaign, but the apparent severe boundary does not transport uniformly. The strongest surviving intervention evidence is that a lightweight received-I/Q path consistently repairs part of the degradation on the same frozen severe-held benchmark at an extreme operating point. The frozen inference controls and a prospectively frozen retrained zeroed-I/Q ablation support attributing the gain to received-I/Q information.

II. Related Work and Claim Boundary

A. Strong and Efficient AMC Representations

CNN and recurrent AMC systems learn directly from I/Q sequences [1], [2], [5]; later work preserves complex structure, fuses temporal and spatial streams, or applies attention [3], [4], [6], [7]. Vehicular and mobile receivers have motivated reparameterized causal convolutions, cross-scale kernel selection with IQ-imbalance and carrier-offset compensation, and channel-robust cumulant features [10]–[12]. Compact and efficient AMC has received sustained attention for cost-sensitive deployment [8], [9]. These lines of work motivate strong I/Q-domain comparators and prevent us from presenting a lightweight architecture or a fusion scheme as the primary novelty.

B. Robust, Cross-Domain, and Interference-Aware AMC

Channel adaptation, masked or contrastive learning, and interference-tolerant features address different nuisance sources [13]–[16], while overlapped-signal classification, denoising masked autoencoding, and cross-domain prototypical adaptation populate the 2024–2026 frontier [17]–[19]. Interference-tolerant recognition [16] and tensor-based malicious-interference recognition for MIMO [20] address structured interference directly. Our setting differs: we neither reconstruct sources nor seek universal interference robustness; we study how the *ranking* of compact spectral versus high-capacity I/Q models changes with the interference condition itself.

C. Distinction: Rank Instability, Transport Boundary, and Repair

Knowledge-guided and multimodal preprints continue these directions [21]–[24]. Ranking changes under distribution shift are a known broader phenomenon: natural shifts can reorder apparent robustness [25], accuracy trends may change across shifted test sets [26], and benchmark suites expose heterogeneous distribution-shift failures [27]. Our contribution is neither the generic possibility of ranking changes under distribution shift, nor another average-dominance claim, nor a universally valid selection surface. The novelty is the controlled wireless-receiver evidence chain that links campaign-level rank instability, an independently generated severe-domain transport test, and a low-cost received-I/Q sidecar repair evaluated on that same held domain. We study ranking instability, use a low-dimensional envelope as a diagnostic hypothesis, test its transfer under independent severe shift, and intervene on the exposed degradation.

III. System Model and VIMD Configuration

A. Mobile Received Signal and Paired Views

We model a receiver-side AMC front end under terrestrial high-mobility conditions. For source sequence $s[n]$ and jammer $u[n]$, one received window is

$$x[n] = \mathcal{R}(\mathcal{H}_s(s)[n] + \kappa\mathcal{H}_u(u)[n] + w[n]) \,, \tag{1}$$

where $\mathcal{H}_s$ and $\mathcal{H}_u$ are independently drawn fading and mobility operators (parameterized by carrier frequency, terminal speed, and 3GPP TR 38.901 [28] TDL-A/C/D profiles during training, with TDL-B/E held out for channel evaluation), κ scales the jammer contribution so that the realized SIR, defined as $10\log_{10}\big(E[|\mathcal{H}_s(s)|^2]/E[|\kappa\mathcal{H}_u(u)|^2]\big)$ on the simulator's tracked powers, matches the target value, w is receiver noise, and $\mathcal{R}$ applies the frozen receiver chain. The carrier frequency is $f_c = 5.9$ GHz, representative of intelligent-transport-system (ITS) and vehicle-to-everything (V2X) operation [29], and the complex baseband sampling rate is 1 MHz. Terminal speeds seen during training are 0, 60, 120, and 150 km/h, giving a maximum nominal Doppler shift of about 820 Hz; the held-speed split uses 180 and 250 km/h, i.e. up to about 1367 Hz, so speed and TDL profile are held out along separate axes. The held-speed split probes deeper intra-window non-stationarity: $T_c \approx 0.423/f_D$ falls to about 0.31 ms at 250 km/h, below the $1024\,\mu$s decision window.

Training pairs two independently impaired observations of the same immutable source; inference uses one view, so pairing is a training and statistical-alignment device, not additional test-time information.

Unanchored in-taxonomy cochannel collisions are excluded: without a physical target anchor, swapping two admissible emitters can preserve the mixture while changing the requested label.

B. Three-Route Spectral Allocation

Let $Z = \mathrm{STFT}(x) \in \mathbb{C}^{F\times T}$. The front end receives real, imaginary, and log-magnitude planes and produces logits A_s, A_j, A_o on the same lattice. The allocation masks are

$$\begin{aligned} &[M_s, M_j, M_o] = \mathrm{softmax}([A_s, A_j, A_o]),\\ &M_s + M_j + M_o = 1. \end{aligned} \tag{2}$$

An environment encoder produces a scalar ambiguous-route gate λ and a bounded bypass ρ. The modulation and jammer routes are

$$Z_m = (M_s + \lambda M_o + \rho) \odot Z, \tag{3}$$

$$Z_j = (M_j + (1-\lambda) M_o) \odot Z. \tag{4}$$

Because these weights are real and nonnegative, every retained coefficient keeps the mixture STFT phase. The bounded bypass ρ is a per-window scalar in the frozen range $[0.05, 0.35]$. It is added only to the modulation route; therefore the retained route weights remain nonnegative but sum to $1+\rho$ rather than one. Because the bypass is added only to the modulation route, the retained routes satisfy $Z_m + Z_j = (1+\rho)Z$ rather than reconstructing the original mixture exactly. This is the ratio-masking construction familiar from supervised source separation [30]–[33]; the elementary phase-preservation property does not imply source identifiability.

C. Simulation-Component Teacher

During training only, independently tracked post-channel target, jammer, and residual components define powers P_s, P_j, P_u on the student's lattice. With $q_k = P_k/(P_s + P_j + P_u)$, the fixed teacher is

$$M_s^\star = [q_s - q_j]_+, \qquad M_j^\star = [q_j - q_s]_+, \tag{5}$$
$$M_o^\star = q_u + 2\min(q_s, q_j). \tag{6}$$

The routes are nonnegative and sum to one; negligible-energy cells are assigned to the uncertainty route by the finite-precision implementation. We call this construction *physics-informed* because the supervision uses simulator-tracked components; it does not imply that the margin teacher or the three-route allocation is individually confirmed (Section V-A). The teacher is not needed at inference.

D. Training Objective

The total loss is

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\text{mod}} + \alpha\mathcal{L}_{\text{teacher}} + \beta\mathcal{L}_{\text{jammer}} + \gamma\mathcal{L}_{\text{link}} \\ + \eta\mathcal{L}_{\text{contrast}} + \theta\mathcal{L}_{\text{orth}},\end{aligned} \tag{7}$$

where $\mathcal{L}_{\text{mod}}$ is the modulation classification loss, $\mathcal{L}_{\text{teacher}}$ is a Jensen–Shannon [34] alignment between student and teacher masks, $\mathcal{L}_{\text{jammer}}$ and $\mathcal{L}_{\text{link}}$ constrain the jammer-family and link-quality auxiliary objectives, $\mathcal{L}_{\text{contrast}}$ is a supervised-contrastive loss [35] over exact-source positives that encourages condition invariance, $\mathcal{L}_{\text{orth}}$ is a route-orthogonality regularizer, and $\alpha, \beta, \gamma, \eta, \theta$ are nonnegative weights. The exact-source contrastive loss is an auxiliary training objective.

E. Information Contract and Complexity

At inference every model receives only the mixture; simulator-tracked target, jammer, and SIR quantities are training-only or post-hoc audit variables. VIMD-Net uses 39500 parameters and 41.8 million reported MACs.

F. Implementation Details

Unless otherwise stated, all models share the same source-disjoint cache, data budget, optimization budget, stopping rule, and validation-only selection criterion; model-specific components follow each variant's predefined configuration.

Input and front end. One example is a complex baseband window of 1024 samples (1024 μs at 1 MHz, 4 samples per symbol, root-raised-cosine roll-off 0.35), power-normalized by the frozen receiver automatic gain control. The spectral front end applies a two-sided STFT with $N_{\text{FFT}} = 64$, hop 16, a periodic Hann window of length 64, and no centering or zero padding, giving a 64×61 time–frequency lattice at 15.6 kHz per bin. The student and the fixed teacher share this lattice exactly.

Architecture width. The sealed configuration uses 24 spectral channels, embedding dimension 48, environment dimension 32, and dropout 0.05, for 39500 parameters. The prospective capacity ladder widens only these three numbers: the M tier uses (40, 80, 48) and the L tier (64, 128, 64).

Optimization and model selection. All fits use AdamW with an initial learning rate of 3×10^{-4}, weight decay 10^{-2}, batch size 16, mixed precision, gradient-norm clipping at 5.0, and a shared 30-epoch budget. Auxiliary losses follow a fixed staged schedule, and the learning rate decays on a cosine to a floor of 0.05 times its initial value once the staged schedule is complete. The reported checkpoint is chosen solely by validation modulation cross-entropy under a common eligibility window and 8-epoch early-stopping rule. Optimizer settings, schedule, and selection rule are identical for every model; no comparator received an architecture-specific sweep, and validation data are never used for evaluation.

Loss weights. In (7), $\alpha = 0.50$, $\beta = 0.25$, $\gamma = 0.05$, $\eta = 0.10$, and $\theta = 0.01$, with label smoothing 0.05 and contrastive temperature 0.12. The route-orthogonality regularizer is part of the A5 bundle but not isolated.

Seeds. The sealed campaign uses 10 algorithm seeds shared across all models, with seed-aligned initialization; matched-subset displays use the first five. Exact schedules, seed identifiers, teacher construction, and cache provenance are recorded in the reproduction package. Fig. 1 illustrates the method.

IV. Evidence Protocol

A. Frozen Source-Disjoint Campaign

The artifact-locked campaign contains 12 model variants, 10 algorithm seeds, 120 fits, 11 evaluation splits, and 100000 training source sequences. Target classes are BPSK, PI/2-BPSK, QPSK, 8PSK, 16QAM, 64QAM, 256QAM, GMSK, CPFSK, and 4FSK, covering PSK, QAM, continuous-phase, and FSK families. Structured jammer families span tone, multitone, chirp, sweep, partial-band, pulse, comb, and out-of-taxonomy (cochannel and OFDM-like) modulated sources. TDL-A/C/D profiles are seen during training, while TDL-B/E profiles support held-channel evaluation. The evaluation grid spans eight SNR levels from -10 to 18 dB. The in-distribution split uses SIR levels -10 to 10 dB in 5 dB steps, whereas the hard-interference split spans SIR $\in \{-15, -10, -5, 0\}$ dB, so SIR $= -15$ dB occurs only in the hard-interference split and SIR $= +5$ and $+10$ dB only in the in-distribution split. Each hard-interference SNR–SIR evaluation cell of Fig. 2 contains 131–181 source-disjoint test windows (mean 156; all ten classes, 5–26 per class), read from the frozen prediction cache. The envelope aggregates of Section V-C use a coarser partition of the same predictions (216–279 source-disjoint windows per held cell, mean 250). The fit-domain division (in-distribution plus hard-interference cells) is fixed in the exploratory analysis protocol before any envelope is fitted. Jammer family, speed, channel, combined OOD, receiver stress, and clean-retention splits share source identities across model comparisons. The receiver-stress split (ADC quantization and per-emitter synchronization) is retained in the artifact package but does not enter any sealed, envelope, or reported selector claim.

B. Comparator Fairness

Table I summarizes the comparators. All models are retrained under the same source-disjoint protocol; "IQFormer-inspired" and "MCLDNN" denote architecture-faithful

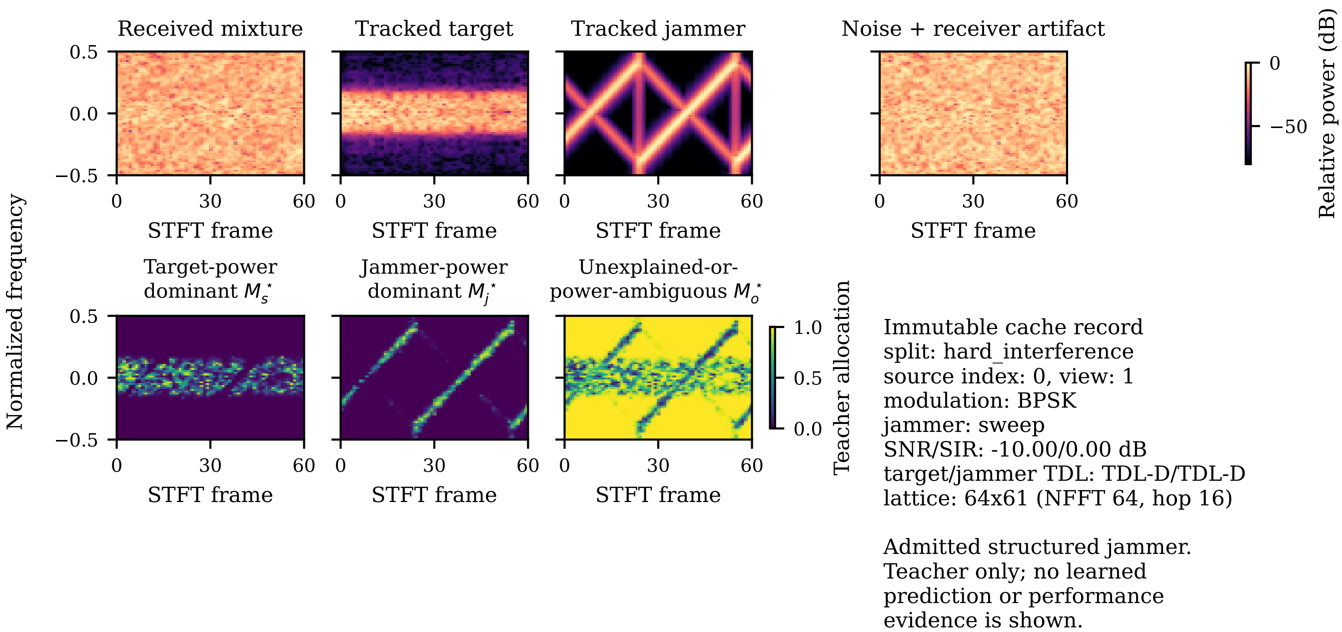

Fig. 1. A deterministic cache record and the fixed simulation-component teacher of Eqs. (5)–(6). Top row: received mixture, tracked target, tracked jammer, and noise-plus-artifact spectrograms; bottom row: the target-dominant, jammer-dominant, and uncertainty route masks. The teacher is a training-only construction from simulator-tracked components; it is unavailable at inference and is not performance evidence.

TABLE I
COMPARATOR AND COMPLEXITY SUMMARY (HARD-INTERFERENCE SETTING).

Model	Domain	Params	MACs	Seeds
A0	spectral	8458	6.3 M	10
A5 / VIMD-Net	spectral allocation	39500	41.8 M	10
sidecar	spectral + received I/Q	46794	43.1 M	10
MCLDNN	I/Q	406070	398.2 M	10
IQFormer-inspired	multimodal I/Q	354984	355.6 M	10
CSSL	I/Q (adaptation)	8631948	155.3 M	10

unified-retraining references, not exact reproductions of every original training recipe. Comparator fairness is enforced at the budget rather than at the tuning level: every model sees the same training source count, the same epoch budget, the same early-stopping patience and checkpoint criterion, and the same validation protocol. CSSL is retained as a supervised adaptation reference; its 8.63 million parameters but only 155.3 million MACs reflect aggressive encoder downsampling and a large projection head, so parameter count alone is a poor cost proxy for this comparator. Figures label this comparator “IQFormer-insp.”; “A5” and “VIMD-Net” denote the same sealed configuration, and “CSSL” abbreviates the contrastive self-supervised learning reference of [15], whose released code snapshot [36] we retrain under our protocol. Because this campaign retrains CSSL under the common supervised budget without executing the released self-supervised pretraining stage, it is reported as a supervised adaptation reference rather than a reproduction of the full released recipe.

As a public-benchmark comparator-fidelity check, the MCLDNN and IQFormer implementations were additionally evaluated on the public RML2016.10a benchmark [1] under the protocol of the official benchmark result table of the IQFormer release [37]: the 11-class set, the full -20 to 18 dB SNR grid, per-stratum $600/200/200$ splits seeded at 233, no augmentation or input-length change. Over three algorithm seeds, all-SNR test accuracy is 58.55% for MCLDNN (seed range 58.1–59.4%) and 65.23% for IQFormer-inspired (64.8–65.8%; seed ranges, not confidence intervals); the reference-release values, averaged over the same twenty SNR bins, are 62.05% and 64.19%. IQFormer-inspired closely matches the reference-release all-SNR accuracy ($+1.04$ pp), whereas MCLDNN remains moderately lower (-3.50 pp) while staying in the same broad all-SNR range. Per-SNR deviations of up to about 8 pp remain, so E1 speaks only to range-level comparator-implementation fidelity.

C. Original Sealed Confirmatory Family

The sealed family contains three hard-interference contrasts: the complete configuration versus A0, the margin teacher versus a proportional-power teacher, and the tri-route model versus a dual-route control. A5 differs from A0 in allocation, teacher, auxiliary objectives, residual bypass, and capacity (about $4.7\times$ parameters), so A5–A0 is a whole-configuration effect.

D. Exploratory and Tier-2 Evidence Classes

All SIR-cell, overlap, fusion, condition-transfer, probe, repair, and capacity analyses are exploratory. In particular, the severe SIR stratification is post hoc. Tier-2 experiments were designed after inspecting the frozen evidence and are outside the frozen confirmatory family; they cannot retroactively change a sealed conclusion. The clean-retention gate was prospectively frozen separately and did not pass; clean behavior is treated as a boundary to explain and repair. The independent severe-domain transfer plan was prospectively frozen before any inference on the new held set. After that evaluation revealed limited cross-regime transport, a second plan prospectively froze the sidecar-repair hypothesis, primary contrast, and decision rule before any sidecar inference on the same held data. A third plan then prospectively froze the inference-only attribution controls of Section V-F before execution. Local plans, hashes, and execution records establish this ordering; no external timestamping service was used. All three remain post-campaign diagnostic evidence and do not alter the sealed family.

E. Statistical Procedures

The sealed family uses a joint non-studentized max-absolute-deviation hierarchical paired bootstrap with 10000 draws, resampling class-stratified source clusters while keeping algorithm seeds as fixed blocks, so pairing is preserved at the source level. The design resolution of 1.31 percentage points is the simultaneous minimum detectable effect at 80% power under this joint interval, an exploratory calibration. Because seeds are fixed blocks, the reported intervals are conditional on the realized seed set; a seed-randomization variant resampling the ten realized seeds with replacement keeps the A5–A0 contrast positive (95% percentile interval $[4.02, 5.19]$ percentage points) while the teacher-form and route-count contrasts span zero ($[-0.15, 0.42]$ and $[-0.15, 1.04]$ percentage points). In the campaign's severe row, all 10 seeds share the same sign; the sign-flip permutation probability 0.0018 is a permutation test over the source-paired samples. The SIR $= -15$ dB aggregate is computed from source-paired predictions pooled across the eight SNR conditions before macro-F1 aggregation; it is therefore not the arithmetic mean of the eight displayed cell-level differences in Fig. 2. The severe-held levels and repair contrasts of Section V-F instead average per-seed macro-F1 values with equally weighted regimes (the seed-wise paired convention); the two conventions are never mixed within a single reported contrast.

V. Results

A. Positive Sealed Whole-Configuration Contrast and Bounded Component Effects

Table II reports the complete sealed family. The complete configuration improves over its shared A0 backbone by 4.57 percentage points, with a simultaneous interval of $[3.65, 5.49]$ percentage points, about 3.5 times the design resolution of 1.31 percentage points. The teacher-form and route-count interventions do not clear zero individually; under the same joint interval their simultaneous upper bounds are 1.08 and 1.36 percentage points, respectively. Thus the configuration-level effect is resolved but cannot be assigned to either isolated component at this resolution, and because the family rule required all three contrasts, the family gate did not pass. Because the A5–A0 interval is part of the same simultaneous family-wise band, its positive whole-configuration contrast remains a valid family-wise-controlled result even though the prespecified joint family gate does not pass. At the same parameter tier, the residual-free A7 variant is nominally above A5 by 0.21 percentage points in the marginal hard-interference aggregate (Fig. 3); treating the ten realized seeds as a random effect gives a 95% percentile interval of $[-0.19, 0.62]$ percentage points, so the difference is not resolved at this design's power; we attribute the compact frontier position to the VIMD family at this budget as a whole.

TABLE II
SEALED CONFIRMATORY HARD-INTERFERENCE FAMILY. DIFFERENCES ARE CANDIDATE MINUS REFERENCE IN MACRO-F1 PERCENTAGE POINTS.

Contrast	Diff.	Sim. low	Sim. high
Full configuration vs A0	4.57	3.65	5.49
Margin vs proportional teacher	0.16	−0.76	1.08
Tri-route vs dual-route	0.44	−0.47	1.36

B. Campaign-Internal Rank Instability across the SNR–SIR Plane

Marginal hard-interference macro-F1 is 0.4406 for A5, 0.4599 for MCLDNN, and 0.4872 for IQFormer-inspired. Those averages favor the larger models. Figure 2 shows why no single campaign-level ranking is adequate: the sign changes with SNR and SIR, including a severe-row reversal. Section V-C summarizes this campaign structure without assuming that it transfers.

At SIR $= -15$ dB, in a post-hoc analysis, A5 gains 9.74 percentage points over MCLDNN ($[7.52, 12.14]$) and 10.46 percentage points over IQFormer-inspired ($[6.78, 14.29]$); these two headline intervals are marginal stratified-bootstrap intervals and are not multiplicity-adjusted. All 10 seeds have the same sign, and the sign-flip permutation probability is 0.0018. Of 64 inspected cells, 12 survive Holm correction; all are on the severe SIR row. These checks stabilize the reversal inside the campaign without converting the post-hoc analysis into confirmation. Section V-F prospectively tests a broad transport interpretation and rejects it on independently generated severe-held data. The reversal against MCLDNN is not explained by a binding training cap: MCLDNN early-stops in all ten seeds with best epochs well inside the shared budget (median 13, range 11–16; Supplement Table VIII).

C. Campaign-Diagnostic Overlap–SIR Envelope

The campaign-internal rank instability motivates a low-dimensional diagnostic surface. Without assuming a universal ranking, the surface summarizes how gain varies with target-support jammer overlap and SIR inside the frozen campaign. For reference b, we fit

$$\Delta_b = \beta_{0,b} + \beta_{1,b} o + \beta_{2,b}\,\mathrm{SIR}, \tag{8}$$

where Δ_b is A5 minus the reference macro-F1 and o is the *target-support jammer overlap*. We avoid the more common "spectral occupancy" because o is not an occupied-bin or occupied-bandwidth fraction. The covariate is defined from the simulator-tracked target and jammer components rather than from the mixture. For each window we compute the

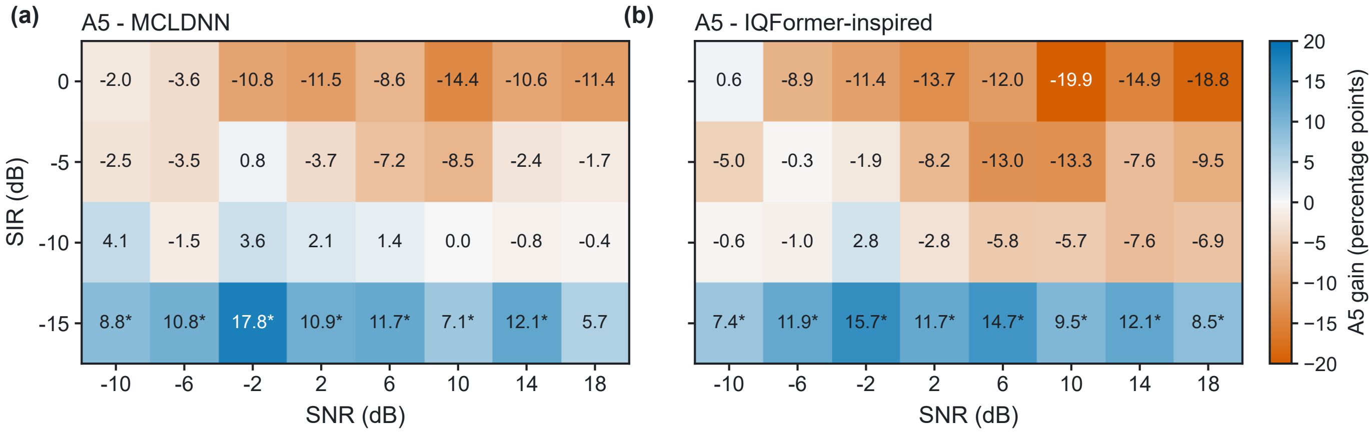

Fig. 2. Campaign-internal post-hoc rank instability over all hard- interference SNR–SIR cells. Each entry is A5 minus the named I/Q reference in macro-F1 percentage points. Blue cells favor A5 and orange cells favor the reference; color and printed values encode the same sign. A star marks a positive unadjusted paired lower interval; Holm-surviving cells are reported in the text.

STFT power of the tracked target and of the tracked jammer separately, order the time–frequency cells by descending target power, and take the target's 99%-energy support $\mathcal{S}$ as the smallest set of cells whose cumulative target power reaches 99% of the total. The overlap is then the fraction of the jammer's total STFT power that falls inside $\mathcal{S}$:

$$o = \frac{\sum_{(f,t)\in\mathcal{S}} P_j(f,t)}{\sum_{(f,t)} P_j(f,t)}. \tag{9}$$

Thus $o = 0.8$ means that 80% of the jammer's energy lies within the target's dominant support, not that the jammer fills 80% of the plane. The cell-level covariate is the mean of o over the windows of that cell.

Equation (9) is evaluated on a fixed analysis lattice, a Hann-windowed 128-point STFT with hop 32, independent of the 64-point VIMD front-end lattice of Section III-F: the front-end lattice is what the model sees, whereas the analysis lattice is a measurement instrument for the physical audit. The overlap is computed once per window when the cache is built, before any model is trained; it is stored in the sealed cache, checksummed in the cache manifest, and read unchanged by every model, split, and comparison in this paper. It is never an input feature of VIMD-Net or of any comparator, and it is not tuned against any model's predictions.

The fit uses 32 aggregates, reconstructible from the protocol. A cell is one (split, SIR stratum, overlap quartile) triple that retains at least 150 windows and all ten classes. The `hard_interference` split contributes 4 SIR strata $(-15, -10, -5, 0\text{ dB}) \times 4$ overlap quartiles $= 16$ cells, and `id_test` contributes 16 further cells over its five SIR strata; its lowest-overlap quartile is dominated by jammer-free windows ($o = 0$, SIR $= 0$ dB cache sentinel) and clears the window floor only at 0 dB, which fixes the per-stratum counts. The primary held set is the 80 finite-SIR interfered cells formed the same way from unseen jammer, unseen speed, held-out channel, and combined OOD (4×5 SIR strata $\times\, 4$ quartiles $= 80$). Clean retention is outside this primary envelope because a jammer-free window has no physical finite SIR. Overlap and SIR are nearly orthogonal across the fitting cells (Pearson $r = -0.003$; -0.03 over all 113 cells), so strong linear collinearity between the two fitted covariates is not evident. We nevertheless interpret Eq. (8) as descriptive rather than causal; low pooled correlation bounds collinearity while leaving confounding open.

Against IQFormer-inspired, the fitted overlap coefficient is 14.38 percentage points per unit overlap and the SIR coefficient is -1.036 percentage points per dB. Magnitude skill is scored against the constant predictor that outputs the mean gain over the fitting cells, evaluated unchanged on the held-regime cells: RMSE 6.23 against 8.37 percentage points ($R^2_{\text{skill}} = 0.45$), and 8.10 against 9.63 percentage points for MCLDNN ($R^2_{\text{skill}} = 0.29$). A covariate ladder isolates each predictor: SIR-only and overlap-only envelopes attain held RMSE 6.84 and 7.46 percentage points against IQFormer-inspired (8.71 and 8.68 percentage points against MCLDNN), so the incremental overlap effect beyond SIR is not separately resolved by the paired bootstrap ($P(\Delta\text{MSE} > 0) = 0.94$ and 0.85, with both paired intervals including zero). Transport tests then bound the surface's role. Leaving out the -15-dB SIR stratum during fitting does not confirm an overlap increment: for MCLDNN the full surface reduces held-out RMSE relative to the constant baseline but predicts the correct sign in only two of four held-out cells, whereas the SIR-only surface preserves all four signs; for IQFormer-inspired the full surface is no better than SIR-only. The independently generated severe-held regimes of Section V-F likewise do not support the surface: envelope sign accuracy there falls below SIR-only and constant predictors. We retain Eq. (8) only as a diagnostic summary of campaign-internal rank variation.

Solving Eq. (8) for zero gain yields break-even overlap references that rise as interference becomes less severe; these are simulator-domain descriptive references conditional on an overlap estimate. The prospectively frozen mechanism expectation was non-increasing gain with overlap; the sealed directional test rejected that sign, and exploratory quintiles instead show monotonically increasing A5–CSSL gain. The positive overlap direction is not specific to this adaptation reference (+19.16 percentage points per unit overlap against

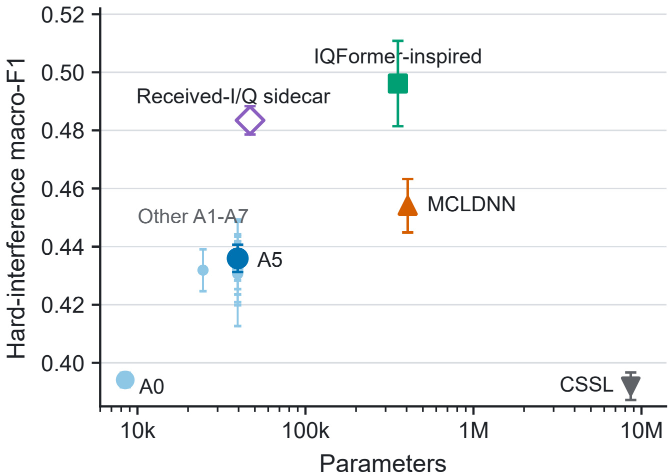

Fig. 3. Hard-interference macro-F1 versus parameter count on the matched five-seed subset. Error bars are ± 1 seed standard deviation. Blue marks denote the spectral family, green IQFormer-inspired, orange MCLDNN, gray CSSL, and purple the received-I/Q sidecar. Filled marks are sealed models; the open sidecar diamond is exploratory and outside the sealed family. The comparison is descriptive and regime-specific.

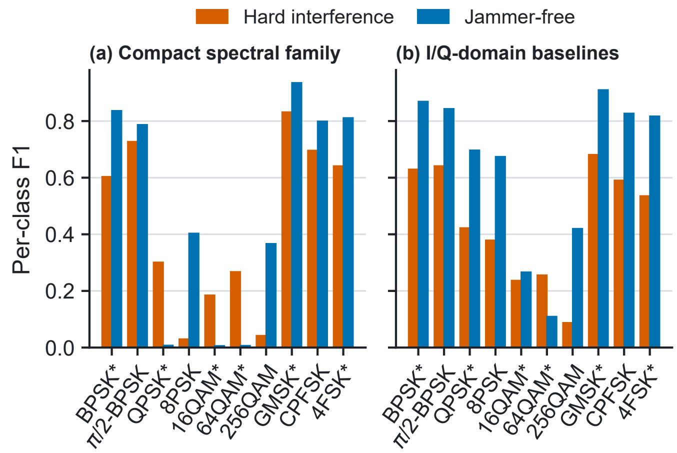

Fig. 4. Clean condition-transfer taxonomy. Bars show family-mean per-class F1 under hard interference (hatched orange) and jammer-free evaluation (blue) for the compact spectral family and I/Q-domain baselines. A star marks a class whose jammer-free condition was absent from training; it is not a significance marker. The same aggregation and scale are used in both panels.

MCLDNN); we return to it in Section VI-B.

D. Complexity and Complementarity

On the campaign hard-interference split, late probability fusion improves over the stronger constituent by 4.71 percentage points with IQFormer-inspired and 5.70 percentage points with MCLDNN. This experiment is a complementarity diagnostic rather than a deployment recommendation: it requires concurrent execution of both constituent models and was not evaluated as the severe-held repair mechanism.

Table I gives the corresponding parameter and MAC budgets; compactness is stated as a parameter and MACs property.

E. Clean-Boundary Diagnosis

The cross-regime severe degradation makes representation robustness central to interpretation. The earlier prospectively frozen clean-boundary diagnosis independently tests whether the compact spectral representation discards information that becomes important under distribution shift. The clean-retention split contains all ten modulations, but the two training views expose jammer-free windows for only 4 classes (PI/2-BPSK, 8PSK, 256QAM, and CPFSK); the remaining 6 classes (BPSK, QPSK, 16QAM, 64QAM, GMSK, and 4FSK) are a condition-transfer test. The four covered classes are marked without a star in Fig. 4, where a star denotes a class whose jammer-free condition was not seen during training. Transfer failure occurs in $27/27$ uncovered compact-spectral class–model cells (100%) versus $1/9$ I/Q-reference cells (11%). The clean-retention gate did not pass. Figure 4 localizes the failure at the class level.

We then followed a prospective decision chain. Frozen-checkpoint linear and MLP probes used a correctly reconstructed jammer-free counterfactual and source-disjoint clean cross-validation; probe performance is the unweighted mean F1 over QPSK, 16QAM, and 64QAM (chance $\approx$ 0.1 on ten classes). The prospectively frozen decision rule declares a representation *information-present, head-limited* when this mean exceeds 0.25 and *representation-limited* otherwise, requiring *all* design–seed cells to exceed the threshold. For A5, five of the six design–seed cells (three algorithm seeds $\times$ the reconstructed jammer-free counterfactual and clean cross-validation probes) lie below the 0.25 threshold; the sole cell above it is borderline at 0.256. Because the decision changes under a 0.20 threshold, we treat the probe result as directional evidence for representation limitation rather than as a threshold-robust classification of mechanism. The strong I/Q baselines exceed the threshold in 18 of 18 cells. The evidence therefore leans toward missing constellation information in the compact spectral representation rather than a classification head that fails to use available information.

Adding clean windows for every class tests the simpler coverage explanation. It changes clean macro-F1 by only 0.15 percentage points ($[-0.31, 0.63]$) and hard macro-F1 by -0.30 percentage points ($[-0.91, 0.27]$). Coverage alone is therefore insufficient. A prospectively frozen negative control gates the modulation route using predicted interference presence. Relative to A5, it changes clean macro-F1 by -0.30 percentage points ($[-0.66, 0.08]$) and hard macro-F1 by -0.18 percentage points ($[-0.72, 0.35]$). The learned gate averages 99.994% on clean windows and 99.991% under hard interference, saturating near the modulated route; this arm serves as a negative control.

F. Representation Repair under Independent Severe Shift

Widening the spectral model without changing its representation domain first separates capacity from representation. The widest spectral tier still trails IQFormer-inspired by 2.84 percentage points on the campaign hard split and by 18.14 percentage points on clean retention. Tested spectral width scaling alone therefore does not remove the representation boundary.

An independently generated severe-held set, frozen before inference, then tests broad transport. SIR is fixed at -15 dB, the same level as the campaign corner cells: R1 combines held jammer families with 180-km/h mobility over training channels, while R2 combines held channels with the same jammers at 250 km/h. Relative to the campaign corner, the held set therefore changes jammer family and speed in R1, and jammer family, speed, and channel in R2. Each regime spans eight SNRs and 6400 source-disjoint examples (12800 total); every checkpoint remains frozen. The campaign-internal corner does not transfer under either aggregation convention. Envelope sign accuracy falls to 0.578 and 0.406, while SIR-only or constant predictors beat the full surface. This experiment therefore establishes a cross-regime transport boundary for both the corner and the -15 dB diagnostic surface.

A per-family split of the held predictions localizes the degradation: under OFDM-like interference every evaluated model sits near the floor (4.60% for A5), while the pulse family remains learnable (29.03% for A5, 55.34% for MCLDNN). The pulse level also explains why MCLDNN's held level exceeds its campaign -15-dB level: the held pulse family yields substantially higher I/Q-domain performance under this evaluation, so out-of-domainness and jammer-family difficulty are confounded in this comparison. Per-family and per-regime levels are tabulated in the Supplement.

The sidecar pools mean, standard deviation, and maximum statistics from a two-channel depthwise-separable received-I/Q branch, fuses its 24-dimensional embedding residually with the spectral path, and adds a side head. It uses only the received mixture and adds about 3% compute over A5. On the original campaign, the same configuration gains 11.77 percentage points on clean retention and 4.37 percentage points on the hard split relative to A5.

On the same frozen severe-held benchmark that exposed the transport failure, the prospectively specified sidecar raises seed-mean pooled macro-F1 from 20.72% to 25.09%, a paired gain of 4.37 percentage points ($[2.96, 5.90]$). Repair is positive in all 10 seeds and all 8 SNR strata. The result is a consistent 4.37 percentage points relative repair at an extreme operating point where every evaluated model remains low in absolute macro-F1, so it should be read as relative robustness repair rather than deployable accuracy. Both severe-held validations remain exploratory post-campaign evidence outside the sealed family. Coincidentally, the campaign hard-split sidecar gain is also 4.37 percentage points, but it is a distinct ten-seed measurement with interval $[3.44, 5.31]$ percentage points.

The frozen controls in Table III separate tested spectral width scaling from received-I/Q information and the added fusion head. Control A supports the negative result: independently trained wider spectral models do not reproduce the repair. Controls B and C are inference-time lesions of the jointly trained sidecar; each lesion removes 2.02–4.12 times the magnitude of the repair effect, and the lesioned variants fall below the unmodified A5 baseline. They establish inference-time dependence on received-I/Q input but cannot quantify the branch's information contribution. The prospectively frozen follow-up retrained the same architecture with the I/Q input permanently zeroed (five matched seeds, identical cache and recipe): the zeroed variant gains -0.54 percentage points versus A5 ($[-1.34, 0.26]$ percentage points), while the matched full sidecar gains 5.39 percentage points$[3.54, 7.66]$ percentage points, and the zeroed variant lands at the A5 level on the campaign hard split, ruling out a training failure. The repair is therefore attributed to received-I/Q information rather than to added capacity or fusion structure.

VI. Discussion

A. What the Rank Instability Establishes

The stable conclusion is not that A5 owns a severe-interference region. Model ranking is condition- and regime-dependent. Marginal scores favor the strong I/Q references, the frozen campaign contains a severe reversal, and an independent regime reverses that result again. This pattern supports condition-bounded model comparison.

B. Why the Diagnostic Envelope Does Not Generalize Uniformly

Equation (8) has skill over 80 held cells inside the frozen campaign, but that skill is heterogeneous by regime. The incremental overlap effect beyond SIR is not separately resolved, and leave--15-dB-out behavior depends on the reference. The prospectively frozen directional hypothesis also fails: gain increases rather than decreases with overlap inside the campaign. The surface remains a diagnostic hypothesis. Its overlap variable additionally requires simulator-tracked target and jammer components and is not directly observable at an uncooperative receiver.

C. Cross-Regime Transport Boundary

The independent severe-held experiment was prospectively frozen before inference and was allowed to reject the campaign interpretation: it covers unseen jammer, mobility, and channel combinations at the same severe SIR level, and its negative result characterizes the boundary of the original finding. It also prevents the in-campaign RMSE and break-even references from being promoted to broad severe-domain transport claims; the boundary is stated without a deployment switching claim.

D. Representation Repair

The intervention chain constrains the explanation while also exposing an important cost asymmetry. Increasing spectral width does not reproduce the severe-held repair, whereas a small received-I/Q branch improves the same frozen benchmark with only about 3% additional MACs over A5. Inference-time lesions establish dependence on the received-I/Q branch, and the prospectively frozen retrained zeroed-I/Q ablation shows that the added capacity and fusion structure alone do not reproduce the gain. The result supports received-I/Q information as the relevant factor within the tested architecture and simulation domain; it does not establish a universal mechanism or deployment-level accuracy.

TABLE III
SEVERE-HELD TRANSPORT, REPAIR, AND FROZEN-INFERENCE CONTROLS. POSITIVE CONTRASTS FAVOR THE ROW MODEL; WHERE REPORTED, CONFIDENCE INTERVALS ARE SEED-WISE PAIRED 95% INTERVALS. THE LESION ROWS ARE INFERENCE-TIME INTERVENTIONS ON THE JOINTLY TRAINED SIDECAR AND MEASURE LESION SENSITIVITY; THE FINAL TWO ROWS (C) REPORT THE PROSPECTIVELY FROZEN RETRAINED ABLATION WITH THE I/Q INPUT PERMANENTLY ZEROED ON FIVE MATCHED SEEDS. SEE SUPPLEMENT.

(a) Full ten-seed severe-held evaluation

Model or intervention	Macro-F1 (%)	Reference	Contrast
A5 spectral baseline	20.72	—	reference level
IQFormer-inspired	23.29	A5	+2.58 percentage points
MCLDNN	33.75	A5	+13.03 percentage points
Received-I/Q sidecar	**25.09**	A5	+4.37 percentage points $[2.96, 5.90]$
Sidecar, I/Q zeroed	12.80	full sidecar	−12.28 percentage points
Sidecar, I/Q shuffled	16.25	full sidecar	−8.84 percentage points
Sidecar, side head removed	7.09	full sidecar	−17.99 percentage points (−13.62 percentage points vs. A5)

(b) Matched five-seed capacity control

Model	A5	Width M	Width L	I/Q sidecar	Seeds
Macro-F1 (%)	20.72	21.77	20.89	**26.11**	5

(c) Prospectively frozen retrained ablation

Sidecar retrained, I/Q permanently zeroed	20.18	A5	−0.54 percentage points $[-1.34, 0.26]$	5
Full sidecar (matched seeds)	**26.11**	A5	+5.39 percentage points $[3.54, 7.66]$	5

E. Vehicular and Learned-Receiver Design Implications

At $f_c = 5.9$ GHz and the held-speed conditions, the channel can evolve materially within the 1024-μs decision window: the nominal coherence time at 180 and 250 km/h is about 42% and 30% of the window, increasing intra-window nonstationarity. Representation robustness is therefore a receiver-design issue as well as a classifier-ranking issue. More generally, the results caution against fixing a learned receiver architecture or switching rule solely from one campaign's marginal leaderboard or condition surface. When compact processing is desirable, preserving a small information-rich signal path may offer a lower-cost robustness mechanism than reverting to a full high-cost I/Q receiver. The sidecar illustrates this tradeoff within the present simulation domain.

VII. LIMITATIONS AND REPRODUCIBILITY

The evidence has explicit limits. First, all results are based on source-disjoint TR 38.901 TDL-profile simulations and do not constitute over-the-air or MAC-/network-layer V2X validation. Second, the sealed family resolves only the A5–A0 whole-configuration contrast; the isolated teacher-form, route-count, and clean-retention gates do not pass. Third, the severe-held regimes change jammer family together with mobility, and R2 also changes channel profile. The transfer result therefore bounds joint transport and does not identify a single causal shift factor. Fourth, IQFormer-inspired and CSSL reach the shared 30-epoch limit in all seeds, so their comparisons retain a budget-sufficiency caveat; MCLDNN early-stops in all seeds and is not subject to the same immediate cap. Fifth, the simulation-component teacher and the overlap covariate require simulator-tracked components that are unavailable in standard public AMC corpora and at an uncooperative receiver, respectively. The RML2016.10a check is therefore limited to comparator-implementation fidelity rather than validation of the proposed representation.

Reproducibility is enforced at the artifact boundary: each manuscript macro stores the source path, row selector, evidence class, precision, and source SHA-256; every figure is rendered from a CSV with its own provenance manifest; and the resolved macros, figure-source CSVs, per-seed aggregates, configurations, and the full provenance manifest are released upon acceptance. The frozen run stores cache identity, checkpoints, predictions, statistics, and execution records, preventing a favorable Tier-2 result from overwriting an unfavorable sealed gate.

VIII. CONCLUSION

Compact spectral and high-capacity I/Q AMC representations do not admit one stable ordering across structured vehicular interference regimes. The sealed campaign resolves a positive A5–A0 whole-configuration contrast but not the isolated teacher or route-count effects, and a post-hoc severe-condition reversal motivates explicit rank-stability analysis. An independently generated severe-domain evaluation then shows that the favorable campaign ranking and the overlap–SIR diagnostic surface do not transport uniformly across unseen regime combinations. On the same frozen severe-held benchmark, a lightweight received-I/Q sidecar provides a consistent relative repair at about 3% additional MACs. Tested spectral width scaling does not reproduce this repair, and a prospectively frozen retrained zeroed-I/Q ablation supports attribution to received-I/Q information rather than to added capacity or fusion structure. Because absolute macro-F1 remains low at this extreme operating point, the result is a robustness repair rather than deployable AMC accuracy. The practical implication is that learned-receiver design should evaluate ranking stability across operating regimes and preserve low-cost complementary signal information when compact representations face domain shift.

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